

Exercise Sheet - 2

STA201: Elements of Statistics & Probability

Question 1

A manufacturer of electronic components is interested in determining the lifetime of a certain type of battery. A sample in hours of life is as follows:

123, 116, 122, 110, 175, 125, 125, 111, 118, 117

- i) Find the sample mean, median and mode.
- ii) Show that the sum of the deviations of each value from the mean is zero.

Question 2

The lifetime of 40 similar car batteries is given in the table below:

Lifetime (in years)	Number of Batteries
1.5–2.0	2
2.0–2.5	1
2.5–3.0	4
3.0–3.5	15
3.5–4.0	10
4.0–4.5	5
4.5–5.0	3

- i) Find the mean, median, and mode of the above distribution.
- ii) Also show the median and mode graphically.

Question 3

The government awarded grants to the agricultural departments of 9 universities to test the yield capabilities of two new varieties of wheat. The yields in kilograms per plot are recorded as follows:

University	1	2	3	4	5	6	7	8	9
Variety 1	38	23	35	41	44	29	37	31	38
Variety 2	45	25	31	38	50	33	36	40	43

Compare the variability in the yield capability using an appropriate measure. Yield capability of which variety shows more consistency?

Question 4

The following data represent the electricity cost for a sample of 50 two-bedroom apartments and a sample of 50 three-bedroom apartments in a city of New Mexico during the month of May last year:

Electricity Cost	Two-Bedroom Apartments	Three-Bedroom Apartments
120–140	2	3
140–160	7	4
160–180	20	10
180–200	10	25
200–220	5	7
220–240	6	1

Compare the variability in the electricity costs using an appropriate measure. Electricity cost of which type of apartment shows more consistency?

Question 5

The following data represent the ages (in years) of 20 randomly selected participants in a training program:

23, 19, 21, 25, 30, 28, 26, 22, 24, 27, 29, 32, 33, 31, 25, 26, 28, 27, 29, 30

- Calculate the first quartile (Q1), third quartile (Q3), and the interquartile range (IQR).
- Find the 40th and 85th percentiles of the data.

Question 6

The following frequency distribution shows the scores obtained by students in a standardized test:

Score Range	Number of Students
40–50	3
50–60	6
60–70	10
70–80	15
80–90	9
90–100	7

- Calculate the first quartile (Q1) and third quartile (Q3).
- Find the 30th and 70th percentiles.

Question 7

The following data represent the daily number of steps walked by 20 individuals over a certain period:

4500, 5200, 4800, 7600, 5000, 4700, 4900, 7200, 5300, 5100, 6800, 7500, 4800, 4900, 4600, 4700, 4950, 5000, 7300, 15000

- i) Identify whether there are outliers.

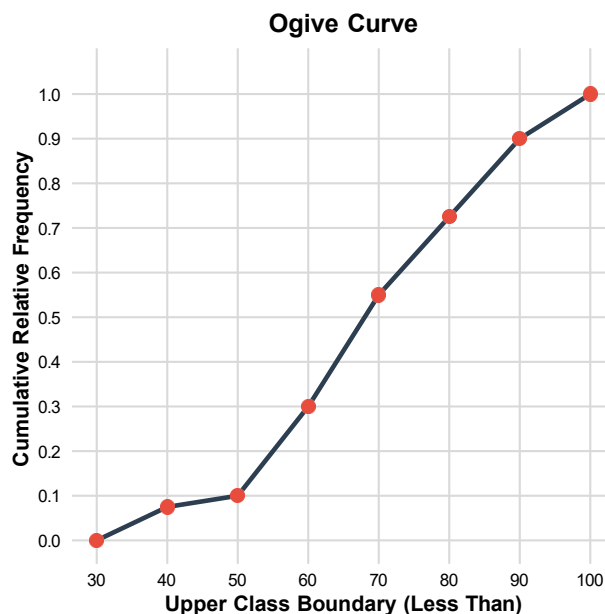
Question 8

Tanzid Hasan and Nazmul Hasan Shanto are both opening batsmen of Bangladesh. Their scores for the last 7 matches are given below:

Tanzid	27, 53, 40, 75, 50, 36, 90
Shanto	39, 61, 84, 25, 120, 40, 104

- a) Find the score above which the top 10% of scores belong for Tanzid.
- b) Suppose you are a shareholder of a BPL team and need to select an opening batsman. Which batsman should you select if you want a more reliable performer? Justify your answer using appropriate statistical measures.

Question 9



An ogive curve representing the distribution of monthly internet usage (in GB) by 100 households is provided in the attached figure.

- a) Estimate the first quartile (Q1), third quartile (Q3), and interquartile range (IQR) from the ogive curve.
- b) Find the approximate values of the 20th percentile and the 85th percentile and interpret the result.

Question 10

The following frequency distribution shows the monthly salaries (in thousand Taka) of employees in a small company:

Salary Range (Tk. in '000)	Number of Employees
10–20	4
20–30	6
30–40	10
40–50	8
50–60	5

- i) Calculate the mean of the salary distribution.
- ii) Find the Mean Absolute Deviation (MAD) from the mean.
- iii) Calculate the variance and standard deviation of the distribution and interpret the results.

Question 11

The following grouped data shows the test scores of students in a competitive examination:

Score Range	Number of Students
300–320	3
320–340	5
340–360	9
360–380	12
380–400	7
400–420	4

- a) Estimate the first quartile (Q1) and the third quartile (Q3).
- b) Use the IQR method to determine the lower and upper fences.
- c) Based on your calculation, would a score of 420 be considered an outlier? Justify your answer.

Question 12

For any dataset arranged in ascending order, determine whether the following statements about quartiles, deciles, and percentiles are **True** or **False**. Justify your answers with reasoning or definitions.

- a) $Q_2 = D_5$
- b) $Q_1 = P_{25}$
- c) $D_7 = P_{70}$
- d) $Q_3 = D_6$
- e) $P_{90} = D_{10}$
- f) $Q_2 = P_{40}$

Question 13

The following data represent the number of hours 10 students studied for their final exam:

4, 6, 8, 5, 7, 9, 6, 10, 5, 7

- i) Find the Mean Absolute Deviation (MAD) of the data.
- ii) Find the variance and standard deviation of the number of study hours.